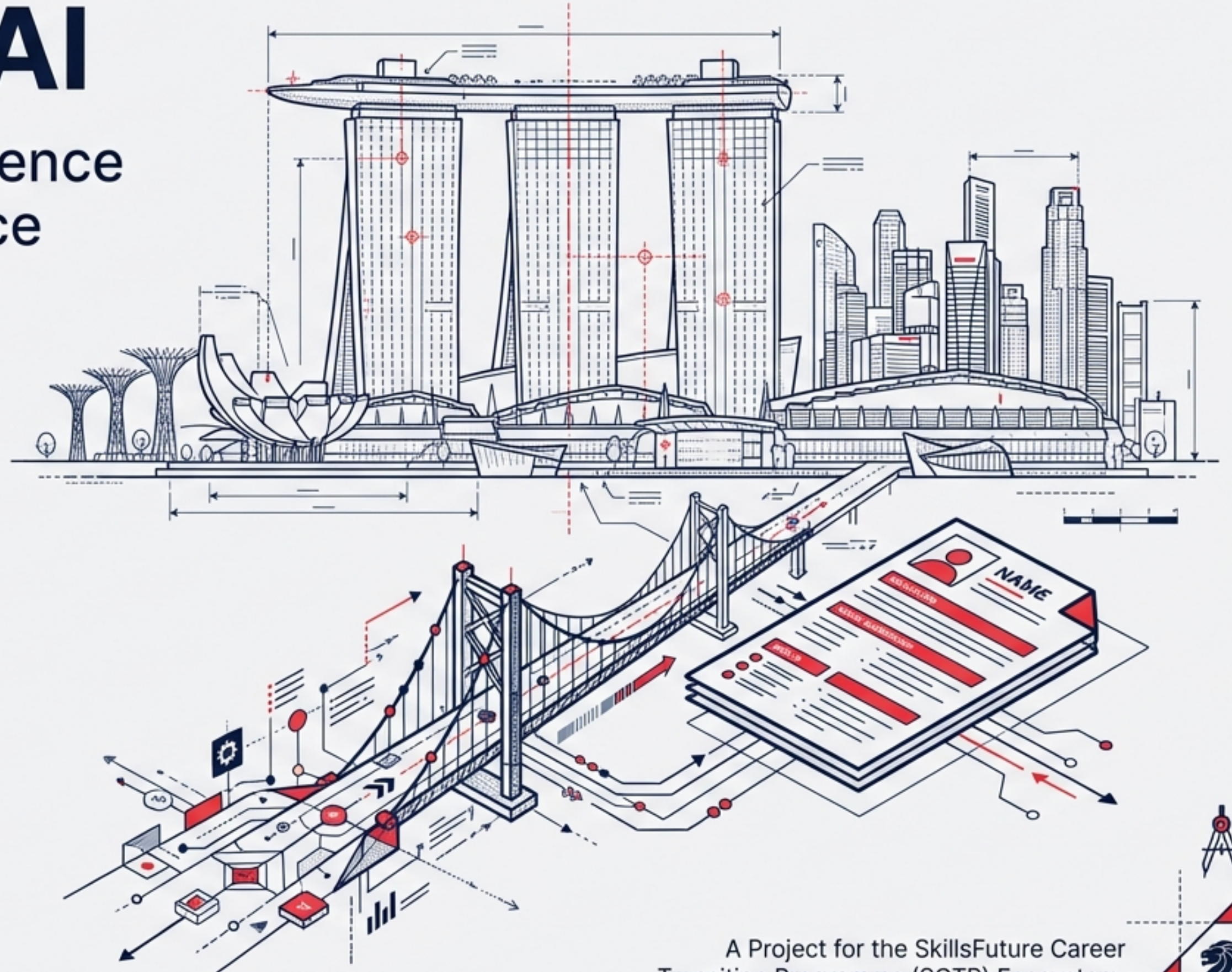


SkillBridge AI

AI-Driven Career Intelligence for Singapore's Workforce

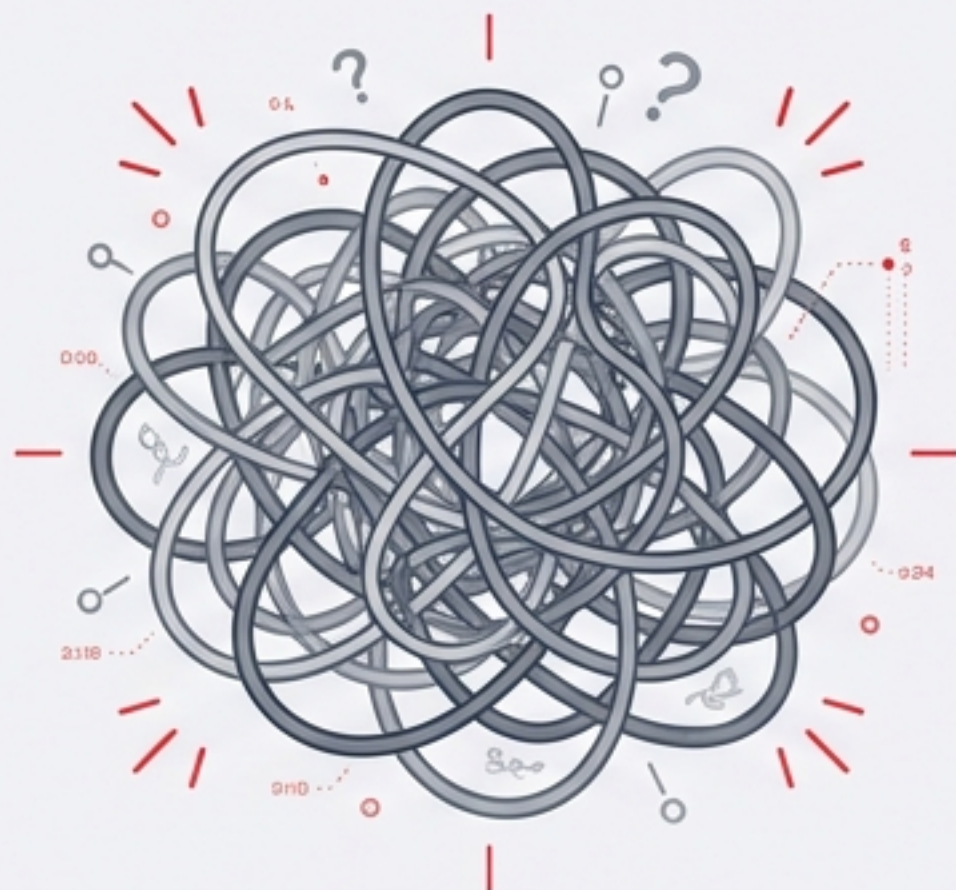
Bridging the gap between current skills and future-ready tech roles.



A Project for the SkillsFuture Career Transition Programme (SCTP) Ecosystem.



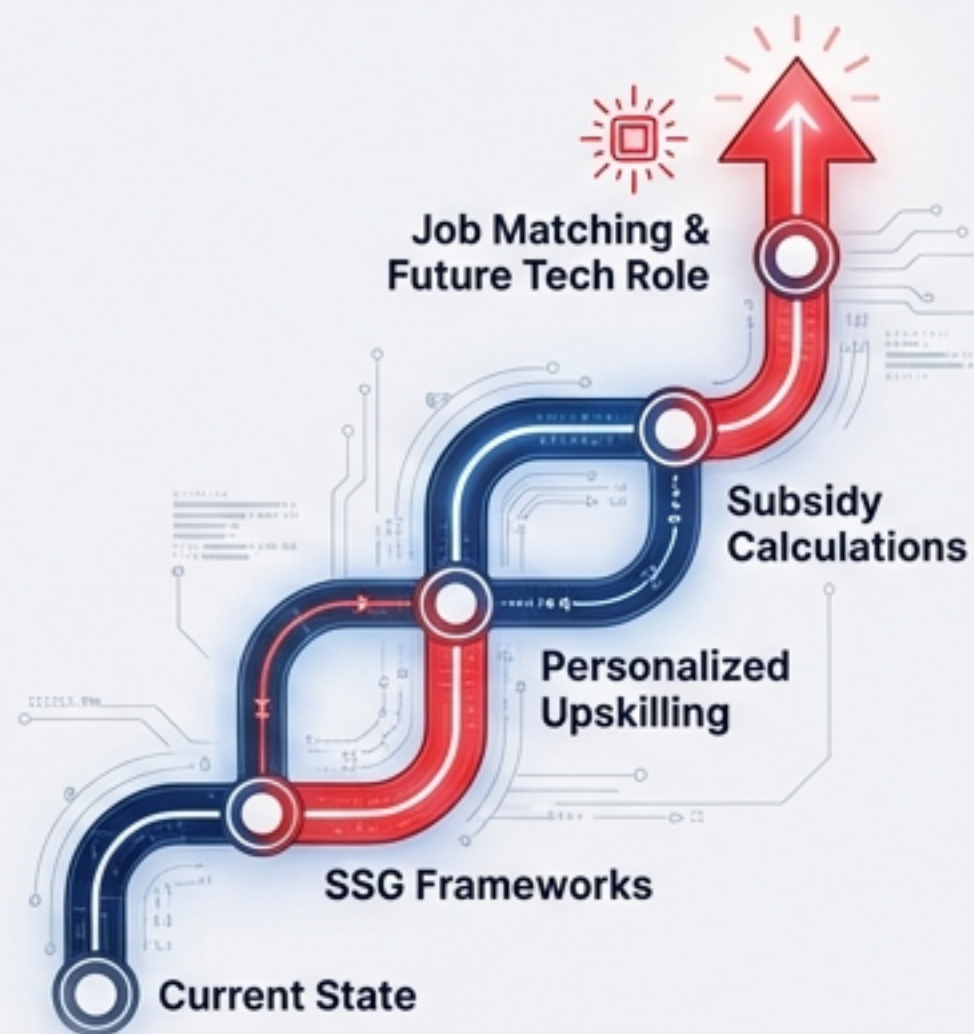
The Problem



The disconnect between existing workforce skills and the rapid evolution of the tech sector creates a navigation crisis.



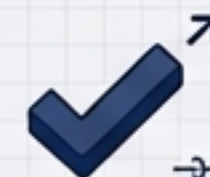
The Solution



A comprehensive platform leveraging SkillsFuture (SSG) frameworks for personalized upskilling, subsidy calculations, and job matching.



Target Audience



SCTP Learners



Mid-career switchers (Age 40+)



Singaporean workforce transitioning to Tech

The Four Pillars of Value



Intelligent Analysis

Smart profile parsing via NLP and Hybrid matching (Content + Rules).



Real-Time Insights

Live market simulation and benchmarks against 2026 growth sectors.



SCTP-validated course pathways and real-time subsidy calculation.

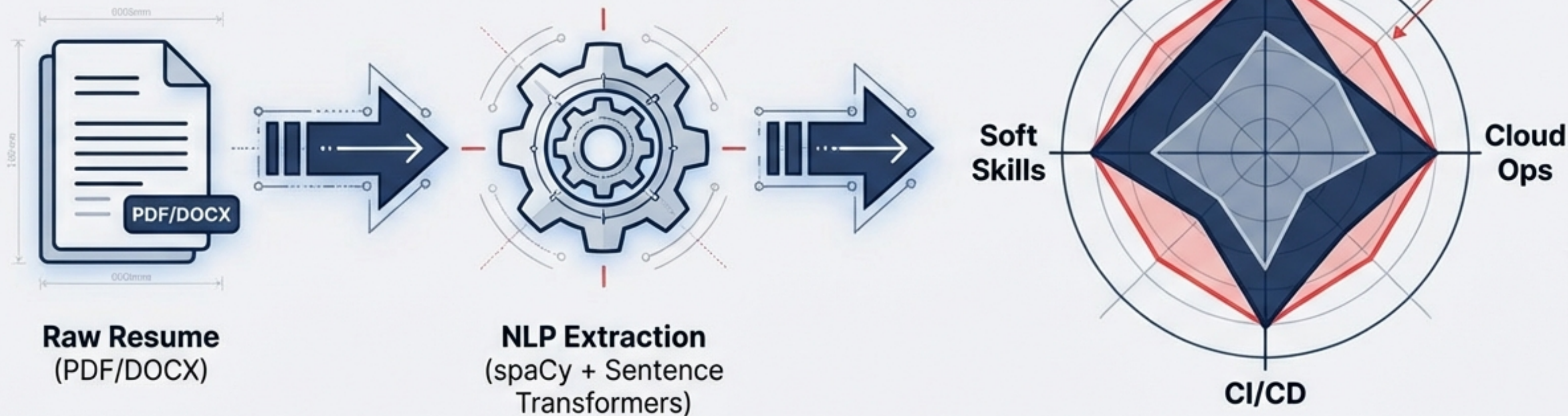


Personalized Learning

Context-aware chat via LLMs and Mock interview simulator.

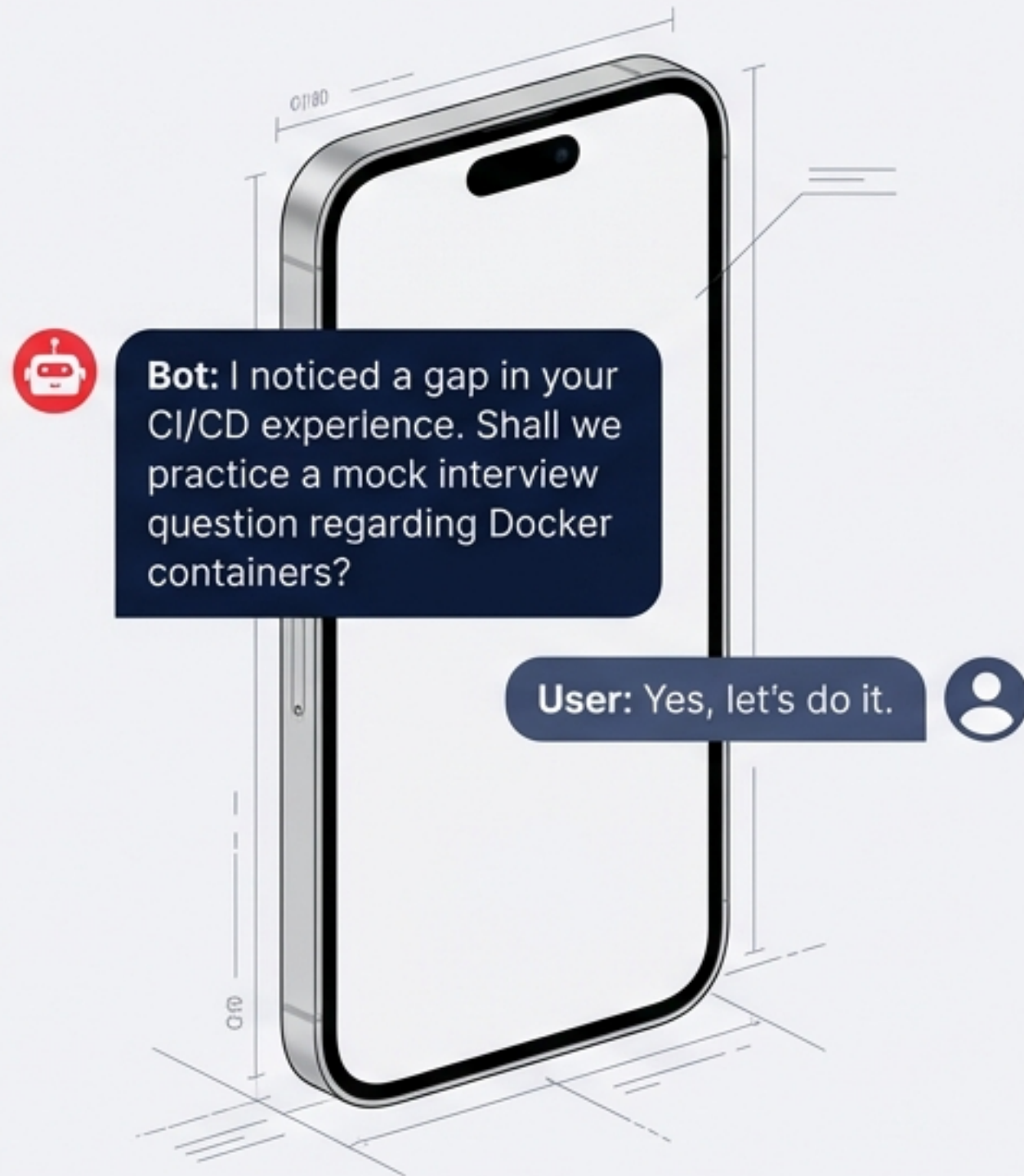
Intelligent Analysis & Hybrid Matching

The Algorithm: 60% Content Similarity + 40% Rule-Based Heuristics



Outcome: Instant visualization of precise skill gaps against target roles.

The AI Coach & Simulator

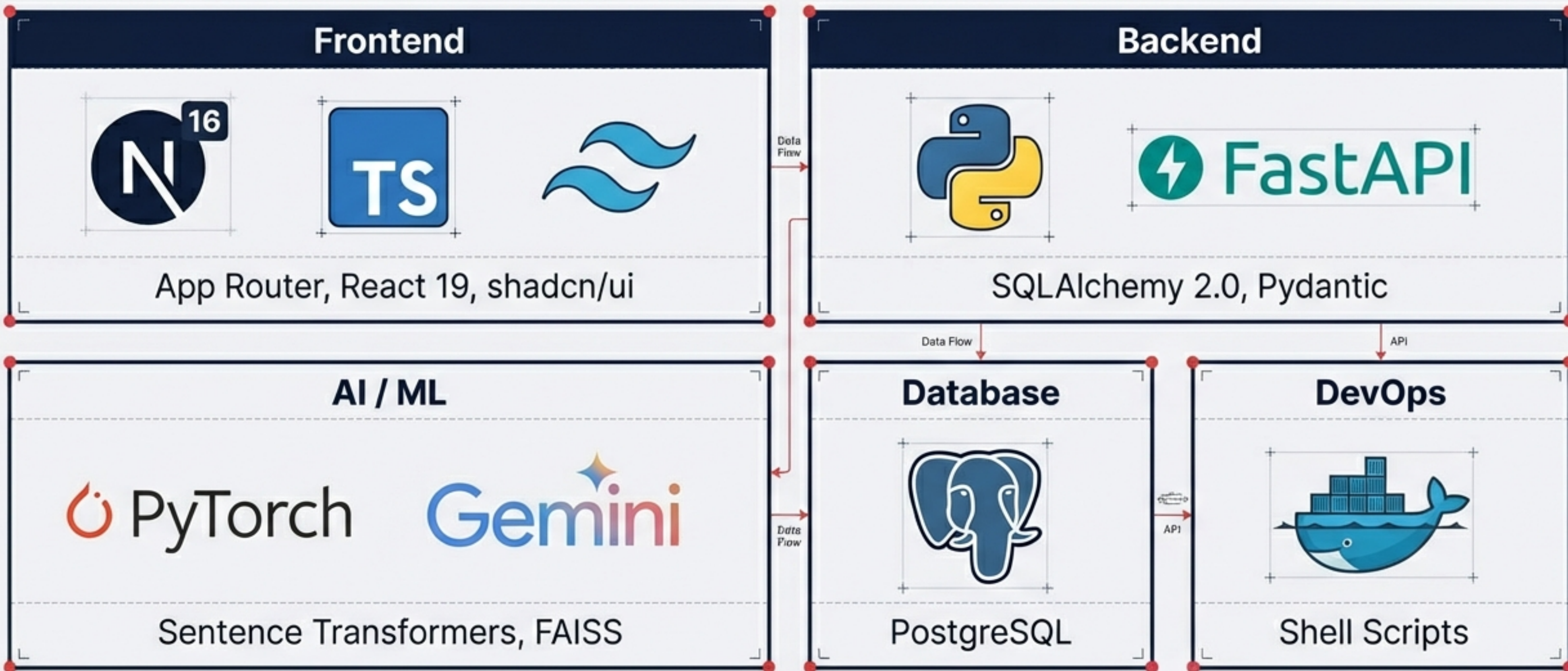


Capabilities

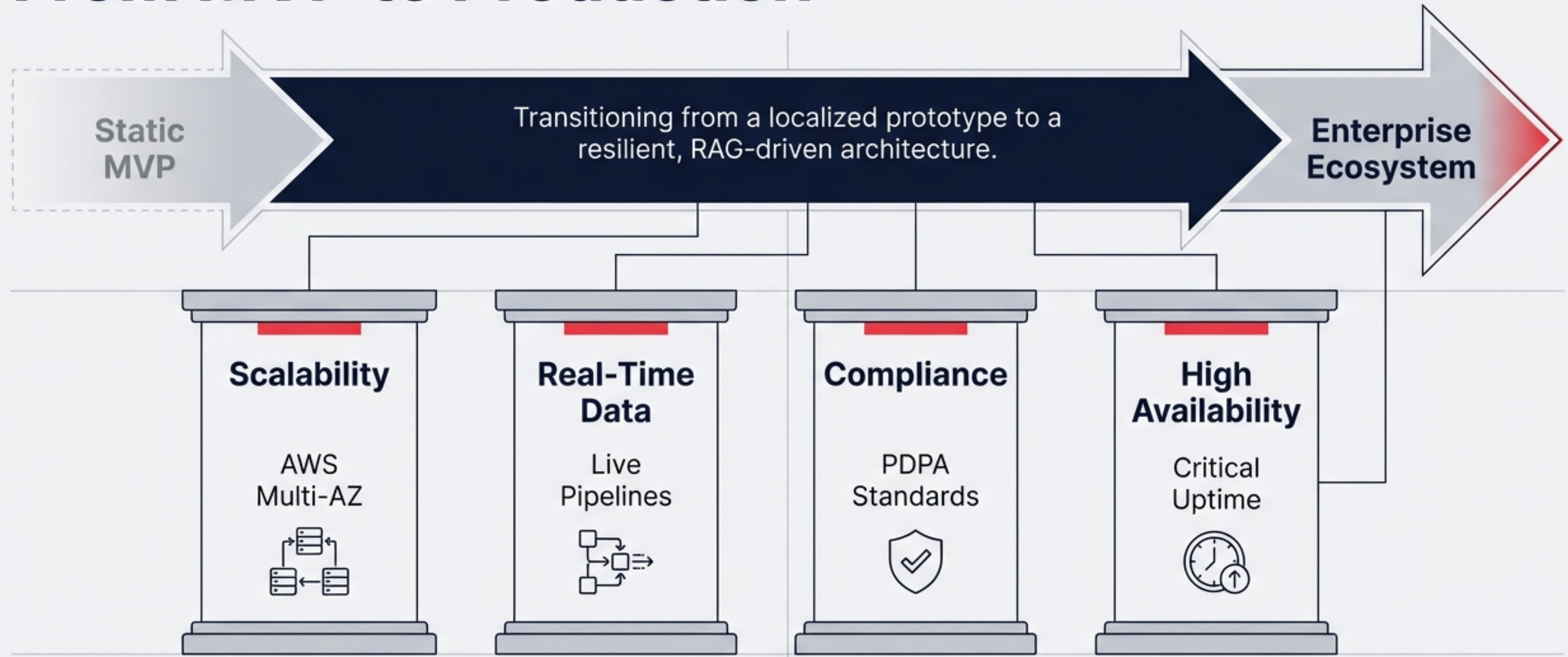
- **1. Context-Aware Chat**
Powered by Gemini, aware of user profile and local trends.
- **2. Mock Interview Simulator**
Generates role-specific questions based on weak points.
- **3. Resume Optimizer**
AI suggestions to rewrite bullet points for impact.



The Foundation: Current MVP Tech Stack



Strategic Vision: From MVP to Production



Data Pipeline Revolution

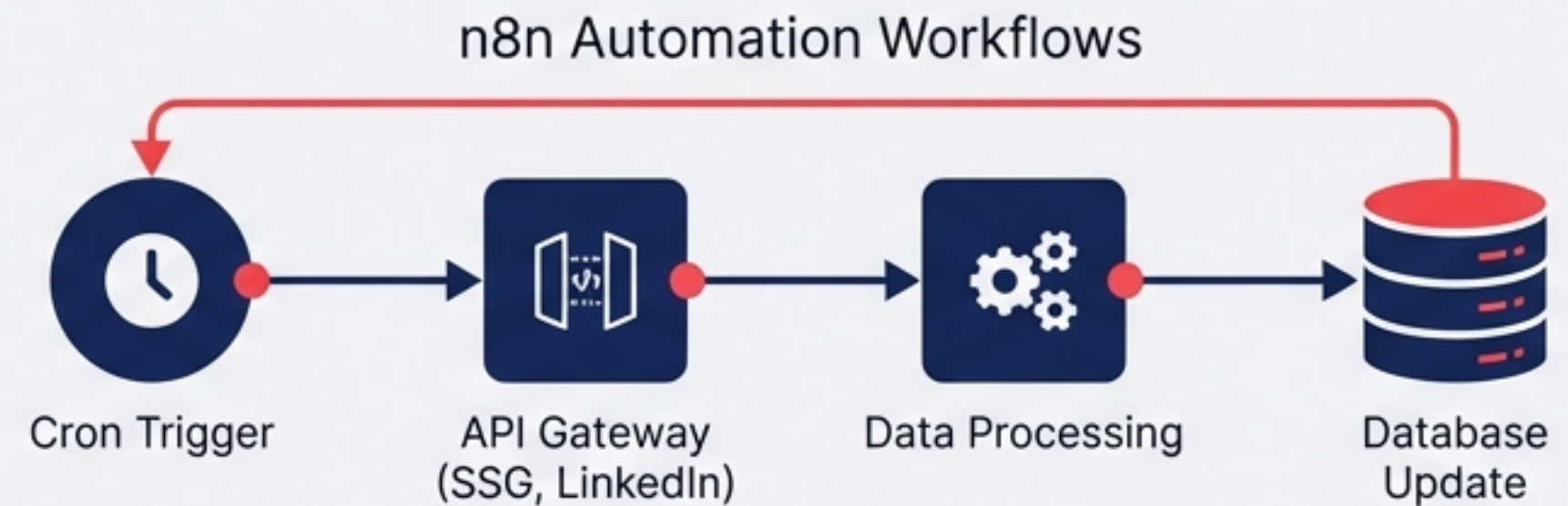
Static vs. Real-Time Ingestion

Before



Manual / Static Snapshots

After

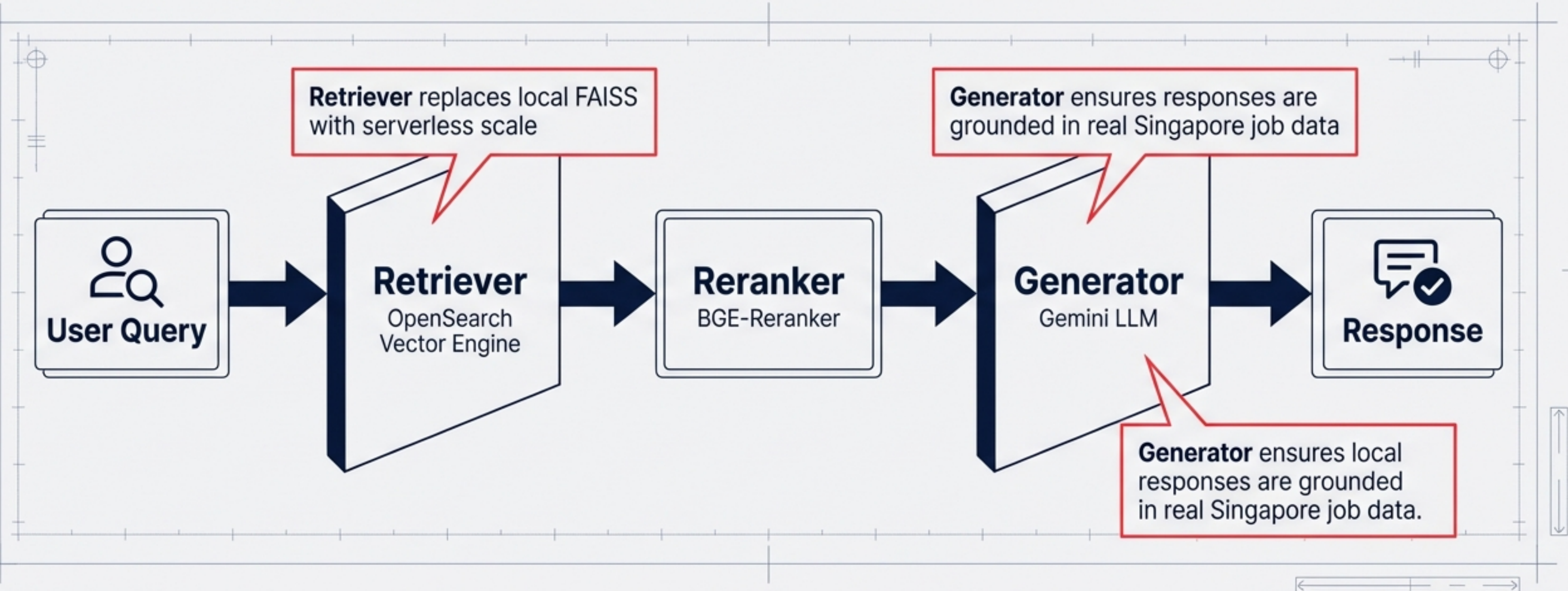


Automated / Real-Time

- Live integration with SkillsFuture & Adzuna APIs.
- Automated tracking of salary trends and hiring volume.
- Ingestion of Skills Framework (SFw) and Technical Skills & Competencies (TSCs).

Intelligence Upgrade: Multi-Agent RAG Architecture

Orchestrating Retrieval-Augmented Generation for Fact-Grounded Answers



Result: Reduced hallucinations and high-fidelity career advice.

Advanced ML & NLP Enhancements

Simulators & Voice

Integration of Whisper / AWS Transcribe for real-time speech interviews

NLP Pipeline

Transformer-based NER for Singapore-specific jargon and acronyms

Embeddings

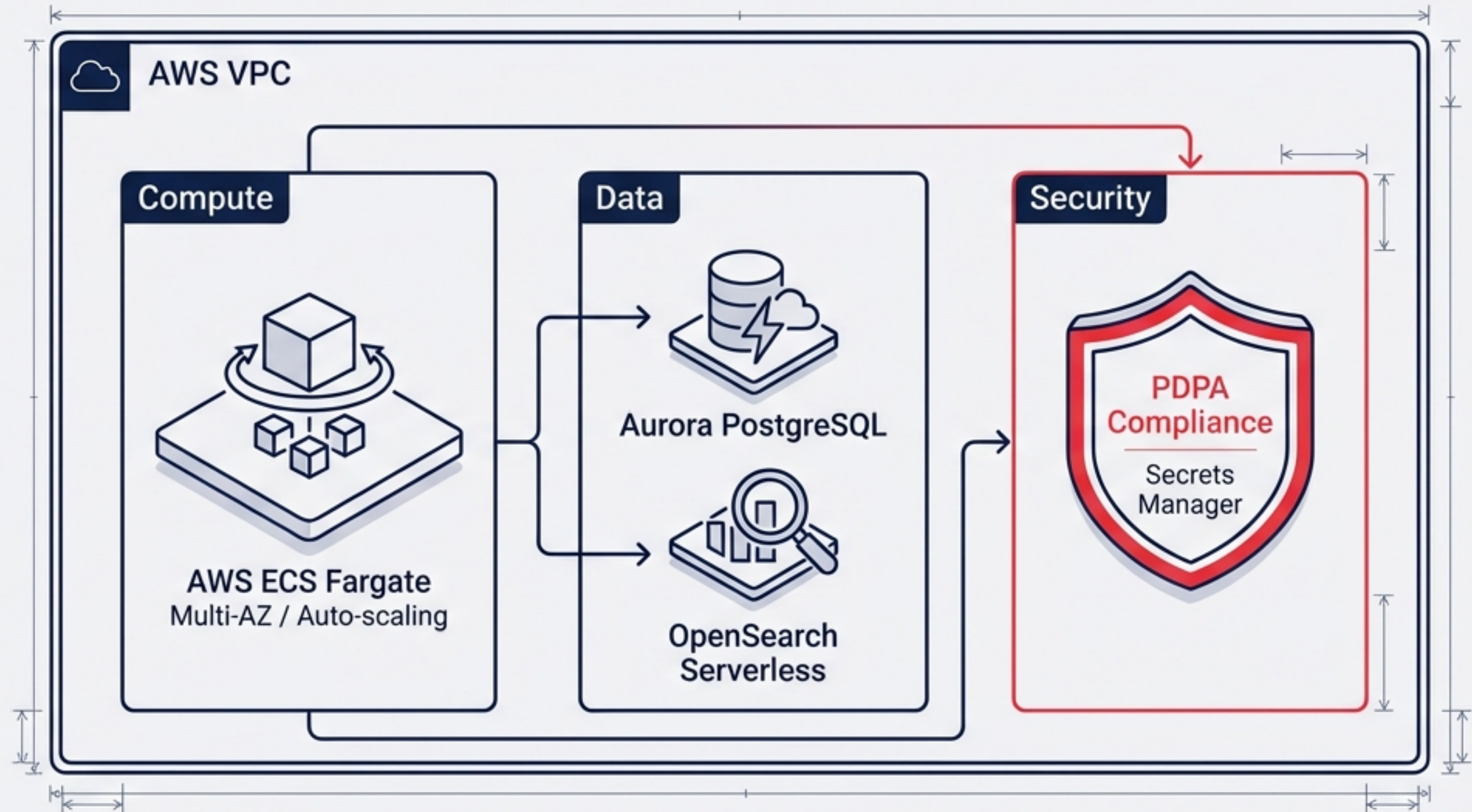
Upgrade to BGE-M3 or OpenAI for deep semantic understanding



Improved
Dimensional
Precision

Cloud Infrastructure & Security (AWS)

- Deployment: **Terraform** (IaC)
- Encryption at rest **and in transit**
- High Availability Architecture



Roadmap Timeline



Financial Engine & Subsidy Logic

Handling MCES and SkillsFuture Complexity

Subsidy Calculator

Course Fee: \$10,000

Age: 42

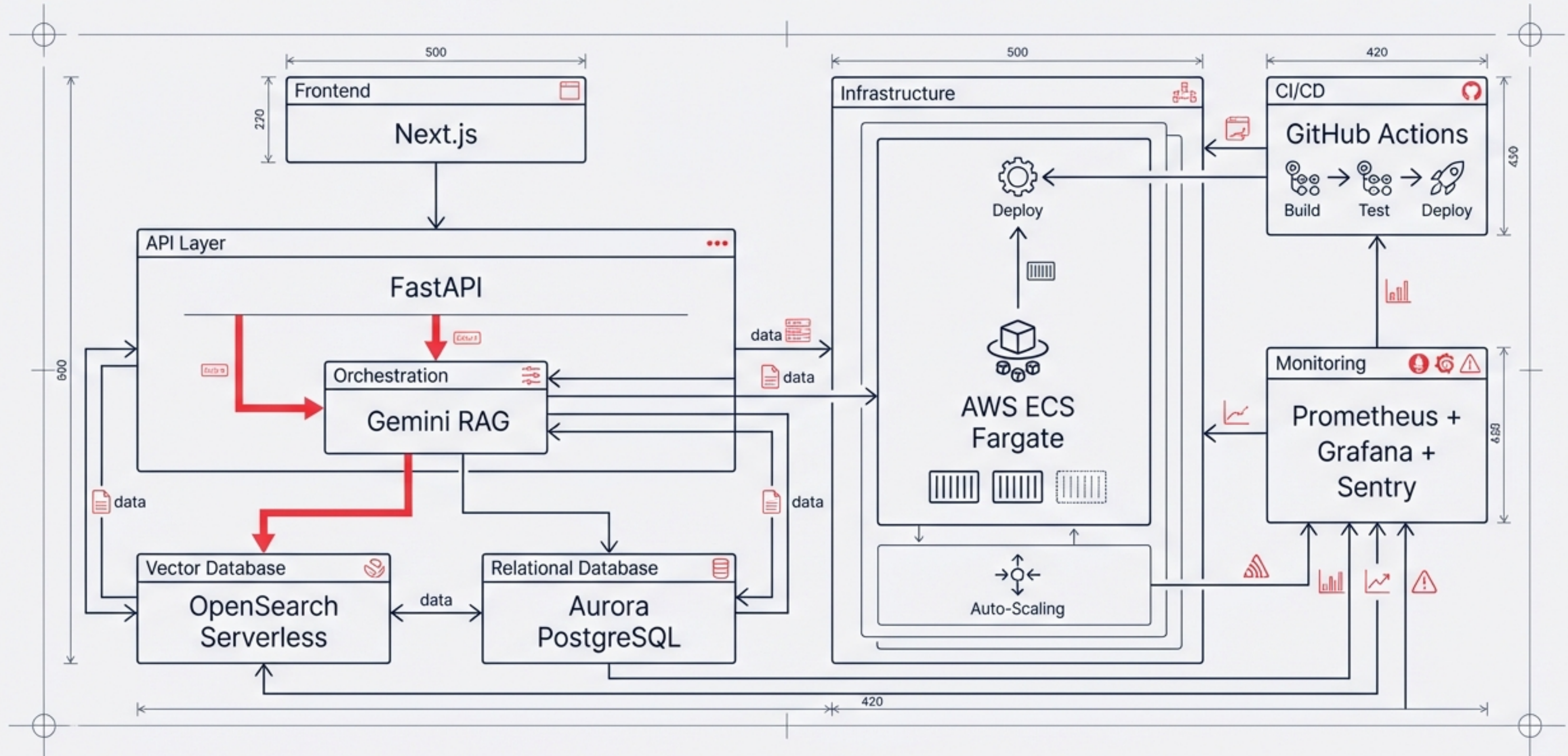
MCES Subsidy Applied (90%): -\$9,000

SkillsFuture Credit: -\$500

Payable Amount: \$500

- **MCES Rules:** 90% coverage for age ≥ 40 .
- **Funding Caps:** Specific to SCTP.
- **Accuracy:** 100% financial calculation logic.

Target End-State Architecture



SkillBridge AI



Empowering Singapore's Workforce

A production-scale platform built for the future of work.

View the Code: github.com/azniosman/dsai-capstone

Review the Technical Roadmap

Distributed under the MIT License.